# A Hybrid AutoML and 3D Docking Workflow for DEL-ASMS-Based Prediction of WDR91 Binders

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## Abstract

A hybrid workflow leveraging AutoML and 3D docking was developed to predict WDR91 binders, using data representations from DNA-encoded libraries (DEL) and affinity selection-mass spectrometry (ASMS). The code and pretrained models are available at https://github.com/koziarskilab/DREAM_Challenge_2025.


## 1. Introduction

We introduce a two-step process to establish robust baselines for compound selection, leveraging automated machine learning (AutoML) and advanced molecular modeling.

Step 1 centers on the use of AutoML with molecular fingerprints, aiming to systematically optimize both classification accuracy and chemical diversity. This step encompasses data preprocessing, optimal metric identification with AutoML, baseline model training, fingerprint correlation analysis, cross-fingerprint model ensembling, and final test set prediction—each component building upon the previous to refine predictive performance.

Step 2 advances this framework by integrating machine learning models with state-of-the-art imbalanced learning techniques, utilizing both molecular fingerprints and SMILES representations. The focus expands to include performance evaluation with imbalanced learning methods, exploration of fingerprint concatenation strategies, and model ensembling. Distinctively, this step incorporates 3D docking analyses for the top predicted compounds, combining machine learning and structural docking predictions to enhance both classification outcomes and chemical diversity in the final selection. Together, these steps provide a comprehensive and systematic approach to compound selection, setting a strong baseline for subsequent innovation and benchmarking in the DREAM Target 2035 Drug Discovery Challenge.

## 2. Methods

Our methodology follows a two-step framework designed for the DREAM Challenge 2025, aiming to establish a strong baseline for compound prioritization against the WDR91 protein target. We depict the framework in Figure 1. All data used throughout the pipeline is sourced from the Artificial Intelligence-Ready CHEmiCal Knowledge Base (AIRCHECK) [1], a curated resource tailored for AI-based molecular discovery. In addition to the training and test sets provided by the challenge organizers, we incorporate an external ASMS validation set from AIRCHECK, which contains 14,764 compounds (4,704 non-binders and 60 binders) to support benchmarking and model selection. Among the 60 active molecules, 30 are publicly available ligands. Note that the publicly available ligands are only used in the Step 2.

### 2.1. Step 1

Step 1 comprises six subcomponents forming a comprehensive pipeline for feature generation, model optimization, and ensemble strategy development.

In Part 1, we begin with the acquisition and preprocessing of molecular data. Each compound is standardized and converted into various structural fingerprint formats to serve as machine-readable input features. The fingerprints used include MACCS, RDKit (RDK), Avalon, and Atom Pair representations, each capturing different levels of molecular structure and connectivity.

Part 2 focuses on metric selection and AutoML configuration. The ultimate goal of this selection is to identify a proxy metric that correlates well with the challenge's evaluation criteria, thereby enabling more effective model tuning even in the absence of a hit-concentrated validation set for the WDR91 target. We utilize FLAML (Fast Lightweight AutoML) (Wang et al., 2021) to explore and optimize classification models across a variety of scoring metrics, including

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[1] https://aircheck.ai/datasets

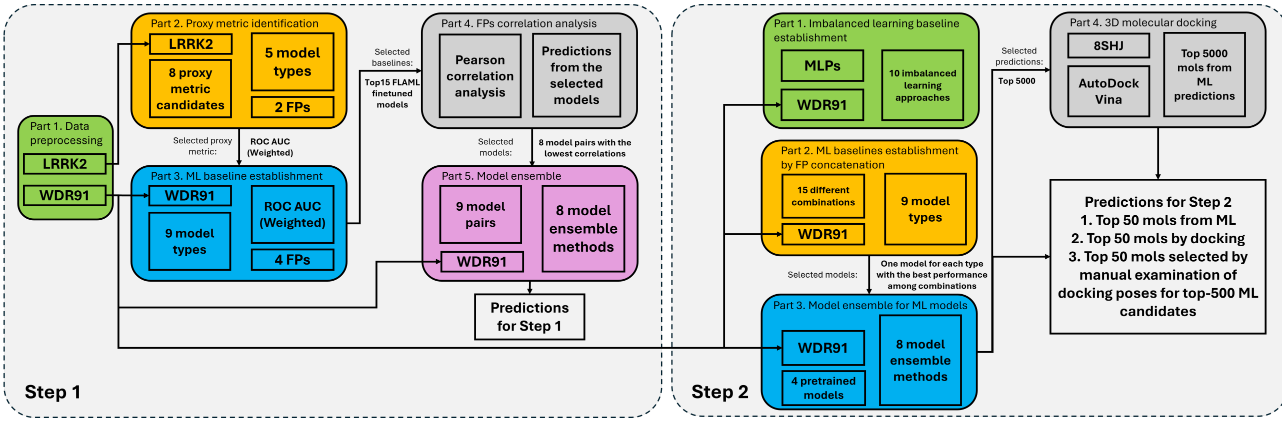


*Figure 1.* Framework of the proposed pipeline.

accuracy, log loss, several F1 score variants, ROC AUC, and average precision. To ensure reliable metric selection, we conduct this evaluation on the LRRK2 dataset, which contains a higher number of hits compared to the WDR91 ASMS dataset. The greater density of positive labels in the LRRK2 dataset provides a more stable basis for comparative metric analysis. Through this process, we identify ROC AUC (weighted) as the most suitable optimization target, as it balances performance across imbalanced classes and better reflects model robustness in real-world screening scenarios.

In Part 3, we train a variety of classifiers using the selected fingerprints. Our modeling suite includes: LightGBM (Ke et al., 2017), standard and depth-limited XGBoost (Chen & Guestrin, 2016), Random Forests (Breiman, 2001), Extra Trees (Geurts et al., 2006), Histogram-based Gradient Boosting (Ke et al., 2017), Logistic Regression with L1 penalty (Wang et al., 2021), CatBoost (Prokhorenkova et al., 2018), and K-Nearest Neighbors (Wang et al., 2021). Each model is trained independently using a single type of fingerprint.

Part 4 addresses the construction of model pairs and the design of ensemble strategies to enhance predictive robustness. We begin by computing pairwise Pearson correlations between prediction outputs from models trained on different molecular fingerprint types. This allows us to identify fingerprint pairs whose models produce weakly correlated outputs, thereby promoting ensemble diversity when combined. This fingerprint-aware pairing reduces redundancy among base learners and serves as a foundation for evaluating a suite of ensemble methods. Specifically, we implement simple averaging, cluster-weighted averaging, majority voting, max voting, min voting, median voting, and two stacking-based methods using logistic regression and random forest as meta-learners. The results are shown in Table 1

Finally, Part 5 consolidates and evaluates the results to identify the most effective combinations of fingerprints and model ensemble algorithms, setting a reproducible baseline for Step 2 of the challenge.

*Table 1.* Average Top 50 cluster-based performance of Step 1 on the validation set. The baseline indicates that without the model ensemble. The best metrics are in bold. The numbers in braces are standard deviations.

| Ensemble method | Hit number | Cluster number | Cluster PRAUC |
|---|---|---|---|
| Baseline (Avalon) | 1.11 (0.93) | 1.11 (0.93) | 0.02 (0.02) |
| Simple averaging | **1.19 (0.46)** | **1.17 (0.42)** | 0.49 (0.39) |
| Cluster-weighted averaging | 1.12 (0.36) | 1.12 (0.36) | **0.66 (0.42)** |
| Majority voting | 0.12 (0.33) | 0.12 (0.33) | 0.00 (0.01) |
| Max voting | 1.18 (0.46) | 1.15 (0.36) | 0.13 (0.06) |
| Min voting | 1.17 (0.56) | 1.15 (0.55) | 0.13 (0.12) |
| Median voting | 0.67 (0.76) | 0.65 (0.72) | 0.04 (0.11) |
| Stacking (logistic) | 1.17 (0.41) | 1.12 (0.33) | 0.23 (0.15) |
| Stacking (random forest) | 0.59 (0.49) | 0.59 (0.49) | 0.01 (0.02) |

## 2.2. Step 2

In Step 2, we aim to build a more advanced and chemically meaningful compound selection strategy by integrating AutoML techniques with state-of-the-art imbalanced learning approaches. While the machine learning models in this step still rely exclusively on molecular fingerprints, we augment the overall pipeline with 3D docking analysis to enhance the biological relevance and chemical diversity of top-ranked candidates. Unlike Step 1, which focuses on baseline modeling, Step 2 systematically explores the interaction between molecular fingerprints and machine learning frameworks under class imbalance conditions. The objective is to maximize predictive performance while ensuring that selected compounds are both structurally diverse and chemically plausible. This step is organized into four interrelated parts that progressively refine the prediction and selection pipeline.

*Table 2.* Average Top 50 cluster-based performance of imbalanced learning baselines among different fingerprint combinations on the validation set. The best metrics are in bold. The best metrics are in bold. The numbers in braces are standard deviations.

| Imbalanced learning method | Hit number | Cluster number | Cluster PRAUC |
|---|---|---|---|
| CE (baseline) | 4.93 (2.89) | 4.07 (2.02) | 0.05 (0.04) |
| LightGBM (ML baseline) | **11.33 (3.77)** | **7.87 (2.03)** | **0.32 (0.05)** |
| CF_B | 4.73 (2.99) | 3.87 (2.20) | 0.06 (0.05) |
| BS | 5.33 (2.85) | 4.27 (2.05) | 0.05 (0.04) |
| CB_CE | 4.87 (2.90) | 4.07 (2.28) | 0.07 (0.06) |
| CS | 4.40 (2.06) | 3.47 (1.55) | 0.05 (0.05) |
| IB | 4.73 (1.91) | 4.07 (1.53) | 0.05 (0.05) |
| CDT | 4.07 (2.66) | 3.27 (1.94) | 0.04 (0.03) |
| Mixup | 5.87 (3.36) | 4.60 (2.13) | 0.09 (0.12) |
| Remix | 4.33 (2.38) | 3.67 (1.91) | 0.12 (0.25) |
| Decoupling | 4.67 (2.61) | 3.87 (1.68) | 0.04 (0.03) |
| BBN | 7.00 (2.65) | 5.93 (2.05) | 0.12 (0.08) |

Part 1 investigates the effectiveness of advanced imbalanced learning strategies when applied to multi-layer perceptrons (MLPs) for compound classification. We include these imbalanced learning methods because the DEL dataset is naturally and severely imbalanced, and we aim to systematically evaluate how classical imbalance-handling techniques affect performance under such conditions. Rather than relying solely on basic resampling techniques, we adopt a suite of algorithm-level imbalance mitigation methods introduced in the ImDrug framework (Li et al., 2022). The evaluated methods include: standard cross-entropy loss (CE), class-balanced focal loss (CB_F) (Cui et al., 2019), balanced softmax (BS) (Ren et al., 2020), class-balanced cross-entropy (CB_CE) (Cui et al., 2019), cost-sensitive loss (CS) (Japkowicz & Stephen, 2002), influence-balanced loss (IB) (Park et al., 2021), class-dependent temperatures (CDT) (Ye et al., 2020), and several augmentation-based techniques such as Mixup (Zhang et al., 2019), Remix (Chou et al., 2020), and Decoupling (Kang et al., 2019). We also assess the Bilateral-Branch Network (BBN) (Zhou et al., 2020), a dual-stream architecture designed for representation and classifier decoupling. These techniques aim to address data imbalance either through reweighting the loss function, augmenting data representations, or decoupling feature and label distributions. Each method is trained with a single type of fingerprint, with performance measured using cluster-based metrics. The average MLPs with imbalanced learning performance among different fingerprint combinations are shown in Table 2. Our findings highlight the sensitivity of MLPs to imbalance and underscore the importance of imbalanced learning, particularly in low-hit-rate datasets like WDR91. However, since SMILES strings are not available for the training set, we are restricted to using MLPs with molecular fingerprints, which limits the model's representational capacity. Although MLPs with imbalanced learning show improvement over their non-imbalanced counterparts, their performance still falls short compared to traditional machine learning models. As a result, we do not incorporate these MLP-based models into the subsequent ensemble stage, which prioritizes higher-performing models trained on diverse fingerprint combinations using more robust algorithms.

Part 2 explores the effects of different fingerprint combinations on model performance. We concatenate various molecular fingerprints, including MACCS, Avalon, Atom Pair, and RDK, to form hybrid descriptors that capture both structural and topological features. By systematically evaluating different fingerprint combinations (15 possible combinations in total), we identify combinations that offer complementary information, resulting in more expressive molecular representations.

Similar to Step 1, Part 3 implements ensemble learning using machine learning models pretrained on different fingerprint combinations. We construct model ensembles by aggregating predictions from diverse base learners, leveraging their unique perspectives on molecular similarity. Voting and averaging strategies are employed to combine model outputs, with ensemble performance measured against individual model baselines. The experimental results on the validation set are shown in Table 3.

*Table 3.* Average Top 50 cluster-based performance of Step 2 on the validation set. The baseline indicates that without the model ensemble. The best metrics are in bold. The numbers in braces are standard deviations.

| Ensemble method | Hit number | Cluster number | Cluster PRAUC |
|---|---|---|---|
| Baseline (Avalon) | 7.50 (3.57) | 5.60 (1.79) | 0.12 (0.12) |
| Simple averaging | **16** | **10** | 0.36 |
| Cluster-weighted averaging | **16** | **10** | 0.36 |
| Majority voting | 3 | 3 | 0.11 |
| Max voting | 15 | **10** | 0.33 |
| Min voting | **16** | 8 | 0.36 |
| Median voting | 14 | 9 | 0.41 |
| Stacking (logistic) | 15 | **10** | 0.37 |
| Stacking (random forest) | 3 | 3 | 0.11 |

Part 4 introduces a structural biology component by performing 3D molecular docking on the top 5000[2] compounds ranked by the best-performing ensemble model. Docking scores are computed to assess potential binding affinity to the WDR91 protein, providing an orthogonal validation signal to the ligand-based prediction models.

For docking, we downloaded seven publicly available protein–ligand crystal complexes from the Protein Data Bank (PDB): 8SHJ, 9DTA, 9DTB, 9EJO, 9EJP, 9MK7, and 8T55. Among them, the ligand in 8T55 binds covalently, while the ligands in the other complexes bind non-covalently; thus, 8T55 was excluded from the study. Docking was performed using the publicly available program AutoDock Vina, with both default and customized scoring functions. Initially, we performed cross-docking to select a generalized protein (i.e., a protein capable of accommodating diverse chemical ligands) for the main docking pipeline. During cross-docking, we tested six docking protocols differing in their scoring strategies to score and rank the docking poses. We started with the default Vina score, then progressed to shape-based (shape_score) and electrostatics-based (esp_score) similarity scoring with six crystal ligands (Bolcato et al., 2022). This was followed by weighted combinations of shape and electrostatics scores (0.4 * shape_score + 0.6 * esp_score, 0.5 * shape_score + 0.5 * esp_score, and 0.6 * shape_score + 0.4 * esp_score) to rank the docked poses.

We selected protein structure 8SHJ along with the scoring function $0.5 \times shape\_score + 0.5 \times esp\_score$ to select the best poses for the top 5000 molecules predicted by the ML pipeline. After docking, we manually analyzed the interactions of the top 500 docked molecules, and selected the ligands that showed interaction patterns similar to those formed by the six crystal ligands. These interactions include hydrogen bonds with THR460, LYS461, ARG462, and THR532, as well as fitting into the hydrophobic pocket composed of LEU465, LEU467, LEU477, and ALA459. The 3D docking results are depicted in Table 4.

Finally, we generate the predictions for step 2 in the following ways:

- We select the top 50 molecules purely based on the prediction scores from ML models;
- We first select the top 5000 molecules based on the prediction scores from ML models, then the top 50 molecules are selected by their docking scores;
- We first select the top 500 molecules based on the prediction scores from ML models, then the top 50 molecules are selected by manual examination of docking poses for the top ML candidates.

[2]The actual number is 4999, we failed to dock the ID_46910.

*Table 4.* 3D docking performance on the validation set. The best metrics are in bold.

| TopN | Method | Hit number | Cluster number |
|---|---|---|---|
| N = 50 | Best ML model | **15** | 10 |
| | docking | **15** | 8 |
| N = 200 | Best ML model | **23** | 12 |
| | docking | **16** | 9 |
| N = 500 | Best ML model | **31** | 14 |
| | docking | **20** | 11 |

## 3. Discussion

### 3.1. Key Finding 1: Model Ensemble Is Essential

Our experiments demonstrate that model ensembling significantly improves performance, particularly in terms of cluster-based evaluation metrics. These improvements are consistently observed in both Steps 1 and 2 of the pipeline (Table 1 and 3). Moreover, ensemble methods effectively balance the trade-off between prediction accuracy and chemical diversity, underscoring their critical role in the late fusion stage of compound selection.

### 3.2. Key Finding 2: Imbalanced Learning Enhances Deep Learning Models

Due to the absence of SMILES strings in the training set, our model choices are limited to traditional machine learning algorithms and MLPs. The results reported in Table 2 highlight that while MLPs with imbalanced learning techniques do not yet match the performance of traditional ML models on cluster-based metrics, some imbalanced learning methods outperform their non-imbalanced counterparts. These results suggest that imbalanced learning strategies hold strong potential for enhancing deep learning models, particularly when full molecular representations such as SMILES strings are available.

### 3.3. Key finding 3: 3D Docking

While molecular docking alone underperforms compared to ML models in terms of predictive accuracy (Table 4), it still demonstrates non-trivial predictive power and offers complementary biochemical insights. In Figure 2, we learn that docking ranks some active compounds favorably that are missed by ML-based methods. This observation motivates the hypothesis that combining ML predictions with docking scores can reduce false positives and incorporate experimental relevance into the selection process. To test this, we evaluate three ensembling strategies that apply different weightings between ML and docking outputs. These strategies aim to strike a balance between statistical learning

| Top50 ML | | | Top 50 3D docking | | |
|---|---|---|---|---|---|
| SMILES | LABEL | CLUSTER_LABEL | SMILES | LABEL | CLUSTER_LABEL |
| COc1c(C#N)cccc1CC(=O)Nc1ccc2c(c1)CCCN2C(=O)c1ccccc1 | 1 | 4 | CN1C(=O)C(c2c(-c3ccc(Cl)cc3)[nH]c3c2c(=O)[nH]c(=O)n3C)c2ccccc21 | 1 | 7 |
| COc1ccc(-c2cncc(C(=O)N[C@H]3CCc4c(C#N)cccc43)c2)c(C)c1 | 1 | 5 | O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc(N2CCC(O)CC2)cc1 | 1 | 6 |
| N#Cc1cccc2c1CC[C@@H]2NC(=O)c1ccc(S(=O)(=O)NCc2ccco2)cc1 | 1 | 14 | O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc(N2CC[C@@H](O)C2)cc1 | 1 | 1 |
| N#Cc1cccc2c1CC[C@@H]2NC(=O)c1ccc(SCc2cn3ccccc3n2)cc1 | 1 | 2 | Cc1cn2cc(C(=O)NC3(c4ccc(Cl)cc4)COC3)sc2n1 | 1 | 11 |
| N#Cc1cccc2c1CC[C@H]2NC(=O)c1ccc(Nc2ncnc3ccccc23)cc1 | 1 | 2 | CN1CCc2cc(C(=O)NC3(c4ccc(Cl)cc4)COC3)sc2C1 | 1 | 9 |
| CC(=O)c1cccc(OCc2cn(-c3ccc(C(=O)N4CCC(C#N)CC4)cc3)nn2)c1 | 1 | 12 | O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc(CN2CCCC2=O)cc1 | 1 | 1 |
| CCN(CC)Cc1cn(Cc2ccc(C(=O)N[C@@H]3CCc4c(C#N)cccc43)cc2)nn1 | 1 | 6 | Nc1ncnc2cc(C(=O)NC3(c4ccc(Cl)cc4)COC3)ccc12 | 1 | 6 |
| NC(CNC(=O)c1cn[nH]c1-c1ccco1)c1cc(F)cc(Cl)c1 | 0 | -1 | O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc(-c2cc[nH]n2)s1 | 1 | 0 |
| CCc1ccc2c(c1)C(=c1sc3n(c1=O)CN(c1ccc(C(=O)O)cc1)CN=3)C(=O)N2 | 0 | -1 | CC(=O)NCc1ccc(C(=O)NC2(c3ccc(Cl)cc3)COC2)cc1 | 1 | 5 |
| O=C(O)c1ccc(N2CN(Cc3ccccc3)CC(c3ccc([N+](=O)[O-])o3)=N2)cc1 | 0 | -1 | O=C(N[C@H](CCO)c1ccc(Cl)cc1)c1ccc(N2CC[C@@H](O)C2)cc1 | 1 | 13 |
| O=C(N[C@H](CCO)c1ccc(Cl)cc1)c1ccc(N2CC[C@@H](O)C2)cc1 | 1 | 13 | NC(=O)[C@@H]1CCCN(c2ccc(C(=O)N[C@@H](c3ccc(Cl)cc3)C3CC3)cc2)C1 | 1 | 13 |
| Oc1nnc(-c2ccco2)c2c1Nc1nnnn1C2c1cccc(F)c1 | 0 | -1 | NC(=O)[C@@H]1CCCN(c2ccc(C(=O)N[C@H](CCO)c3ccc(Cl)cc3)cc2)C1 | 1 | 13 |
| Cn1ccnc1C(Nc1c(F)cc(C#N)cc1F)c1ccc(F)cc1 | 0 | -1 | O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc2snnc2c1 | 1 | 0 |
| COCc1cc(C)nc(NN=Cc2ccc(-c3ccccc3C(=O)O)o2)c1C#N | 0 | -1 | O=C(COC1CCCNC1)Nc1cccc2c(O)cccc12 | 0 | -1 |
| Cn1nc(CN(C(=O)c2ccc(F)cc2)c2ccccc2)c2c1CCN(S(C)(=O)=O)C2 | 0 | -1 | O=C(NCCN1CCOCC1)c1cn2ccccc2n1 | 0 | -1 |
| CN(CC(=O)N1N=C(c2ccco2)CC1c1ccco1)Cc1nc2ccccc2c(=O)[nH]1 | 0 | -1 | CC(=O)NC(C)c1ccc(S(=O)(=O)Nc2ccc(F)cc2C#N)s1 | 0 | -1 |
| N#Cc1cccc(C2CC(c3cccc(O)c3)=NN2c2ncccn2)c1 | 0 | -1 | O=C(N[C@H](CCO)c1ccc(Cl)cc1)c1ccc(CN2CCCC2=O)cc1 | 1 | 1 |
| Cc1cccc(-c2c3c(=O)n(C)c(=O)n(C)c3c3n2CCOC3c2ccc(O)cc2O)c1 | 0 | -1 | CCOc1ncccc1CNC(=O)C1CCN(C(C)=O)CC1 | 0 | -1 |
| CN1C(=O)C(c2c(-c3ccc(Cl)cc3)[nH]c3c2c(=O)[nH]c(=O)n3C)c2ccccc21 | 1 | 7 | O=S(=O)(NCc1ccccc1)N1CCN(C2C=CCCCC2)CC1 | 0 | -1 |
| CN1C(=O)[C@H](c2c(-c3ccc(Cl)cc3)[nH]c3c2c(=O)[nH]c(=O)n3C)c2ccccc21 | 1 | 7 | COc1ncccc1CNC(=O)C1CCN(C(C)=O)CC1 | 0 | -1 |
| Cc1nn(-c2ccccc2)c2c1C(c1ccco1)c1c(ncn3nc(-c4cccnc4)nc13)O2 | 0 | -1 | COc1ccc(N2CCC(C(=O)NC3CCCCCCC3)CC2)nn1 | 0 | -1 |
| N=C1SC(=Cc2ccc(-c3ccc(Cl)cc3C(=O)O)o2)C(=O)N1c1nccs1 | 0 | -1 | O=C1NC(Cc2nc(-c3ccc(N4CCOCC4)cc3)no2)c2ccccc21 | 0 | -1 |
| N#Cc1c(N2C(=O)Nc3ccccc3N2O)sc2c1CCCC2 | 0 | -1 | CCNC(=O)C(C)NS(=O)(=O)c1ccc(OC(C)C)nc1 | 0 | -1 |
| NC(=O)[C@@H]1CCCN(c2ccc(C(=O)N[C@@H](c3ccc(Cl)cc3)C3CC3)cc2)C1 | 1 | 13 | O=C(CCc1ncc(-c2ccccc2F)o1)Nc1cc(C2CCCOC2)[nH]n1 | 0 | -1 |
| Cc1ccc(-c2n[nH]c3c2C(c2ccccc2)N(c2ccc(C(=O)O)cc2)C3=O)cc1 | 0 | -1 | Cc1nnnn1-c1ccc(C(=O)NC(C)COc2ccc(F)cc2F)cc1 | 0 | -1 |
| O=C1CC(c2cc(F)cc(F)c2)Cc2nc(N3CCc4ccccc43)ncc21 | 0 | -1 | COc1ccc(CNC(=O)N2CCC(c3nnc4ccccn34)CC2)cc1 | 0 | -1 |
| CC(NC(=O)c1cc[nH]n1)c1ccc(Oc2cccnc2)c(F)c1 | 0 | -1 | O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc(C2(O)COC2)cc1 | 1 | 5 |
| O=C(c1ccc(N2CC[C@@H](O)C2)cc1)N1CC(O)(c2ccc(Cl)cc2)C1 | 1 | 1 | CC(C(=O)NCc1cn2ccccc2n1)n1c(=O)oc2ccccc21 | 0 | -1 |
| COc1cc(C2CC(=O)Nc3c2sc(C(=O)O)c3-c2ccc(Cl)cc2)ccc1O | 0 | -1 | Cc1ccc(S(=O)(=O)NN=Cc2ccc(N(C)C)cc2)cc1 | 0 | -1 |
| O=C(O)c1cc(-c2ccc(C=NN3C(=O)C4C5C=CC(CC5)C4C3=O)o2)ccc1Cl | 0 | -1 | O=C(c1cn(C[C@@H]2CCCN2Cc2ccncc2)nn1)N1CCc2ccccc2C1 | 0 | -1 |
| CN(C)c1ccc(C(=O)N(C)C(C(=O)O)c2cc(Cl)cc(Cl)c2)cc1 | 0 | -1 | NC(=O)c1ccc(NC(=O)c2ccncc2)cc1 | 0 | -1 |
| Cc1ccn(CC(=O)N2CCn3nc(C(=O)N4CCc5ccccc5C4)cc3C2)n1 | 0 | -1 | O=C(NCc1cccnc1)c1ccc(N2CCCS2(=O)=O)cc1 | 0 | -1 |
| O=C(Nc1ccc(N2CCc3sccc3C2)nc1)C1CCCN(C(=O)c2ccco2)C1 | 0 | -1 | O=C1CCC(C(=O)N2CC(OCc3cccc4ccccc34)C2)N1 | 0 | -1 |
| COC(=O)c1ccc([C@@H]2[C@@H](C(=O)O)c3ccccc3C(=O)N2OCc2ccccc2)cc1 | 0 | -1 | CC(CC(=O)NC/C=C/CNC(=O)c1cccc2[nH]c(=O)[nH]c12)NC(N)=O | 0 | -1 |
| O=C(Nn1ccc2c(nnc3c(-c4ccccc4)cnn32)c1=O)c1cccnc1 | 0 | -1 | Cc1ccn(-c2ccc(C(=O)NC(C)Cc3c[nH]cn3)cc2)n1 | 0 | -1 |
| COc1cc(Cl)ccc1CC(NC(=O)c1cccc2ccnn12)C(=O)O | 0 | -1 | CC(CNC(=O)c1nc(=O)[nH][nH]1)Cc1cccs1 | 0 | -1 |
| O=c1ccc(-c2ccc(S(=O)(=O)Nc3ccc(-c4ccno4)cc3)s2)n[nH]1 | 0 | -1 | NCc1ccc(C(=O)NC2CCCNc3ccccc32)o1 | 0 | -1 |
| CCOC(=O)c1ccc(N2C(=O)c3[nH]nc(-c4ccco4)c3C2c2cccs2)cc1 | 0 | -1 | COc1ccccc1-c1[nH]c2[nH]c(=O)[nH]c(=O)c2c1C1C(=O)N(C)c2ccccc21 | 0 | -1 |
| CCC(C(=O)NC(C)c1ccccc1)n1nc(C)n2c(cc3sccc32)c1=O | 0 | -1 | CCOC(=O)c1nn[nH]c1C1CCCN1C(=O)COCC1CCC1 | 0 | -1 |
| O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc(N2CC[C@H](O)C2)cc1 | 1 | 1 | COc1ccc(C2=NC(=Cc3cc(OC)c4c(c3OC)OCO4)C(=O)O2)cc1 | 0 | -1 |
| O=C(NC1(c2ccc(Cl)cc2)COC1)c1ccc(N2CC[C@@H](O)C2)cc1 | 1 | 1 | O=C(NCc1ccc(F)cc1)c1cn2c(n1)COCC2 | 0 | -1 |
| O=C1c2ccccc2NC(c2cccn2-c2cccc([N+](=O)[O-])c2)N1O | 0 | -1 | c1ccc(Cn2nnnc2CN2CCC(CNc3ccccn3)C2)cc1 | 0 | -1 |
| CC(=O)N1N=C(c2cccs2)CC1c1cccc2cccnc12 | 1 | 16 | CCCC(=O)N1CCC(C(=O)Nc2cccnc2)CC1 | 0 | -1 |
| Cc1c(Cl)cnc(NC2CCCN(c3cccnc3)C2=O)c1C#N | 0 | -1 | O=C(CCC(=O)N1CCN(c2ccccn2)CC1)NC1CCCC1 | 0 | -1 |
| C=CCN1C(=O)C(=C2C(=O)Nc3ccccc32)SC1=Nc1ccc(C(=O)O)cc1 | 0 | -1 | O=C(CCC(=O)N1CCN(c2ccccn2)CC1)NC1CCCC1 | 0 | -1 |
| Cc1noc(C=Cc2ccc(-c3ccccc3C#N)o2)c1[N+](=O)[O-] | 0 | -1 | O=C(NC1CCN(C(=O)C2CCCCC2)CC1)C1Cc2cn[nH]c2CN1 | 0 | -1 |
| CC(O)(Cn1ccccc1=O)C(=O)NCc1ccc(N2CCCS2(=O)=O)cc1 | 0 | -1 | COc1ccccc1CNC(=O)c1cnccn1 | 0 | -1 |
| Cn1nc(C(=O)NC2CS(=O)(=O)Cc3ccccc32)cc1OC(F)F | 0 | -1 | O=C(NCC(c1ccccc1Cl)N1CCCC1)c1cnc2[nH]c(=O)[nH]c2c1 | 0 | -1 |
| CCc1cnc(CN(CC2Cc3ccccc3O2)c2cc(C#N)ncn2)o1 | 1 | 14 | COc1ccccc1-c1[nH]c2c(c1C1C(=O)N(C)c3ccccc31)c(=O)n(C)c(=O)n2C | 0 | -1 |
| COc1ccc(C2=NOC3C4CC(C23)C2C(=O)N(c3ccccc3)C(=O)C42)cc1 | 0 | -1 | CCn1ccc(C(=O)NCC2CCOCC2)n1 | 0 | -1 |

*Figure 2.* Comparison of the top 50 molecules between ML models and 3D docking.

and structural plausibility.

## 4. Conclusion

This work introduces a hybrid AutoML and 3D docking pipeline for predicting WDR91 binders from DEL-ASMS data, with a focus on achieving both predictive accuracy and chemical diversity. Among all datasets used, the external ASMS validation set from AIRCHECK was particularly informative for benchmarking model generalization and refining ensemble strategies. Our results underscore the value of model ensembling, which consistently improved performance across both pipeline stages. We also explored imbalanced learning techniques in MLPs, which improved classification performance on low-hit-rate datasets. While these methods did not outperform traditional machine learning models in this setting, they revealed promising directions for handling imbalance in future deep learning frameworks. Integrating 3D docking into the workflow may further enhance compound prioritization by providing structural validation for top-ranked molecules.

One limitation encountered was the scarcity of confirmed binders, which constrained model training and evaluation. Additionally, the absence of SMILES strings limited the exploration of sequence-based deep models. Future work will focus on expanding training data, incorporating richer molecular representations, and systematically evaluating imbalanced learning across different architectures.

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